

Full-length Article

Temporally-regulated genetic access to IL-1 β -expressing cellular networks in homeostasis and following peripheral or central immune stimuli

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ABSTRACT

Background: The proinflammatory cytokine Interleukin-1 beta (IL-1 β) regulates nearly all aspects of immune function. In the brain, IL-1 β is implicated in neural and immune functions under both basal and inflammatory conditions. Under basal conditions, IL-1 β is known to alter sleep, memory, and affect. Under inflammatory conditions, IL-1 β can induce sickness behaviors, HPA activation, and exacerbate neurological and psychological disorders. Sensitive detection and specific manipulation of IL-1 β -expressing cells in the brain is currently not achievable; therefore, we generated the first mouse line to allow both robust visualization and genetic manipulation of the IL-1 β -expressing cells.

Methods: The IL-1 β -TRAP mouse was generated through Crispr9-mediated recombination. IRES-CreER^{T2} was inserted following exon 7 after the stop codon of the *Il1b* gene which yielded an IL-1 β -IRES-Cre-ER^{T2} mouse (IL-1 β -TRAP). To visualize IL-1 β -expressing cells, IL-1 β -IRES-Cre-ER^{T2} line was crossed with ROSA26-lox-stop-lox-tdTomato mouse to generate the IL-1 β -IRES-Cre-ER^{T2}:ROSA26-lox-stop-lox-tdTomato (IL-1 β -TRAP-reporter). The IL-1 β -TRAP-reporter mice were given intraperitoneal (i.p.) Tamoxifen before peripheral LPS or central IL-1 β administration. PBS was used as control. The IL-1 β -TRAP-reporter mice were also exposed to repetitive closed head injury (CHI) or i.p. kainic acid. tdTomato was allowed to express for 7d and distribution of IL-1 β -expressing cells were observed via immunohistochemistry and/or lightsheet microscopy.

Results: Under basal conditions, IL-1 β was found primarily to be expressed in MHCII⁺ and CD206⁺ meningeal and ventricular macrophages with sparse IL-1 β -expressing microglia and neurons in brain parenchyma. Following i.p. LPS and i.c.v. IL-1 β injections, IL-1 β expression was found in meningeal macrophages and parenchymal P2YR12⁺ microglia. Following CHI, IL-1 β -expressing macrophages increased in the meninges and IL-1 β -expressing microglia were induced in the parenchyma at the injury site and along white matter tracts.

Conclusions: This is the first time active populations of cytokine-expressing cells are visualized, characterized, and genetically accessed using a mouse line with a knockin CreER. With this tool, we identified the choroid plexus as the predominant IL-1 β -expressing region in homeostasis. Further, this mouse can be used to identify newly activated IL-1 β -expressing cells; they can be targeted for manipulation in physiological and pathological contexts.

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1. Introduction

Complex cellular and molecular interactions between the nervous and immune systems are common to numerous disorders of the central nervous system. Elevated levels of pro-inflammatory cytokines have been correlated with the onset and severity of multiple mood, affective, and cognitive disorders (Dantzer et al., 1999; Yirmiya et al., 2000; Zhang et al., 2023). Current methods to study the cellular sources of these cytokines, however, are not sensitive enough to localize cells expressing low levels of cytokines and lack the ability to specifically manipulate them *in vivo*. Therefore, there is a need to create a method for the sensitive detection and manipulation of cytokine-producing cells.

One crucial cytokine, Interleukin-1 beta (IL-1 β), is a potent pro-inflammatory cytokine known as a “master inflammatory cytokine” due its ability to regulate multiple neuroimmune activities in both homeostasis and disease (Basu et al., 2004). During homeostasis, IL-1 β is known to regulate sleep patterns (Baracchi and Opp, 2008; Opp et al., 1991); learning (Depino et al., 2004); and memory (Jones et al., 2018; Spulber et al., 2009). During infection, injury, or disease, IL-1 β is produced to orchestrate CNS-controlled neuroimmune defenses including fever (Cartmell et al., 1999), increased slow wave sleep (Shoham et al., 1987), sickness behaviors (Liu et al., 2019); and increased HPA-axis activity (Matsuwaki et al., 2014). In chronic pathological conditions, changes to affect, mood, presence of autoimmune activity, or neurodegeneration are coupled with elevated IL-1 β levels (Dantzer et al., 1999; Capuron and Dantzer, 2003). Blockade of IL-1 β activity as a therapeutic for neurological diseases, however, has led to mixed results in clinical trials, possibly due to unintended consequences of blocking the beneficial physiological actions of IL-1 β . It's long known that extremely small amounts of IL-1 β are sufficient to trigger powerful IL-1 β -induced effects (Sims and Dower, 1994). As such, picograms/mL levels of IL-1 β were detected in the CSF in the normal brain by biochemical and immunological assays; yet changes of IL-1 β at this low level regulate sleep and learning (Krueger et al., 1998). It is critical, therefore, to precisely describe where and when IL-1 β is produced and regulated under physiological and pathological conditions. Current methods of detecting IL-1 β expression such as *in situ* hybridization and immunohistochemistry, fail to detect IL-1 β -expressing cells in the normal brain (Shimada and Hasegawa Ishii, 2021; Quan et al., 1998; Quan et al., 1999); however pharmacological inhibition or genetic deletion of IL-1 β or IL-1 Receptor (IL-1R1) leads to blunted homeostatic functions such as memory formation (Spulber et al., 2009) or proper stress responses (DiSabato et al., 2020; Goodman et al., 2024), suggesting low levels of brain IL-1 β must be produced in homeostasis. Therefore, there is a need for more sensitive techniques to detect IL-1 β -expressing cells and to modify these cells to determine their impact on CNS function.

One salient question is: *What are the cellular sources of IL-1 β under different conditions?* After peripheral or central immune challenges or chronic psychological stress, where IL-1 β is greatly elevated, various studies have detected brain IL-1 β expression in different cells including endothelial cells (Proescholdt et al., 2002), microglia (McKim et al., 2018), choroid plexus cells (Shimada and Hasegawa Ishii, 2021), astrocytes (Jones et al., 2018; Garber et al., 2018), oligodendrocytes (Vela, 2002), and even neurons (Hallett et al., 2010). It is entirely unclear which of these cellular sources of IL-1 β are verifiable and whether they causally contribute to IL-1 β -linked CNS functions.

Recent advancements in the genetic toolbox include the development of tamoxifen-inducible conditional knockouts, enabling cell-type specific and temporal control of gene knockouts, installing cellular reporters, and creating modifiable cells. This was achieved with the generation of a fusion protein of Cre recombinase fused to cytosolic restricted Estrogen Receptor (CreER^{T2}). Upon binding to an estrogen analog, tamoxifen or the tamoxifen metabolite 4-OHT, conformational changes allow Cre recombinase to enter the nucleus of a cell thus restricting Cre-mediated recombination in a temporal manner. From this technique, a key innovation in neuroscience was the creation of the

cFOS-CreER^{T2} mouse (cFOS-TRAP), which enabled genetic access to active neurons during predetermined time windows (Guenther et al., 2013). This technique, which “traps” specific cell populations linked to various neurological functions, has yet to be applied to other systems, such as the immune system.

Here, we report the creation of the first tamoxifen-inducible mouse line to track IL-1 β expression, which we call the IL-1 β -TRAP mouse. Mating of our IL-1 β -TRAP mouse to a ROSA26-LSL-tdTomato reporter mouse creates an IL-1 β -TRAP-reporter mouse that allows for the permanent visualization of IL-1 β -expressing cells in the CNS during homeostasis or following various peripheral/central inflammatory challenges. This mouse can then be used to identify the contribution of the specific, temporally restricted, subsets of IL-1 β -expressing cells in physiological and pathological contexts.

2. Materials and methods

2.1. Gene targeting strategy

To express iCreER^{T2} together with *Il1b*, the IRES2-iCreER^{T2} was knocked into the 3' of exon 7 of *Il1b* after the stop codon. The 5' and 3' homologous arms for homologous recombination (HR) span about 1.4 KB (see Fig. 1A). To achieve that purpose, the embryonic stem cell (ESC)/HR procedure (proprietary technology developed by Biocytogen), was used to establish the IL-1 β -IRES- Cre-ER^{T2} KI allele. The candidate sgRNAs were selected to target the 3' of exon 7 of *Il1b*. In terms of specificity, they were screened by the CRISPR design tool (<https://www.sanger.ac.uk/htgt/wge/>). The selected RNAs were then determined according to on target activity use UCATM (Universal CRISPR Activity Assay), which is a sgRNA activity detection system developed by Biocytogen. The Cas9, sgRNAs and targeting vector were microinjected into zygotes, which were subsequently transplanted into pseudopregnant mice. Cas9/sgrNA and zygote injection techniques were used to produce targeted KI mice. Due to the rapid cleavage of embryonic stem cells during early embryonic development, the F0 generation mice were genetic chimeras. Thus, the identification of the F0 mice based on their tails DNA only suggests a kind of genotype, which does not certainly indicate a heritable genotype. Hence, a heritable genotype was further determined by verification of genetic analysis in the F1 mice. Chimeric offspring were crossed with wild-type C57BL/6 mice to produce the F1 generation. The germ line transmission was then confirmed by Southern blot and PCR of tail DNA from mice of the F1 generation.

2.2. Southern blot analysis

The restriction enzymes BglII (IRES2 Probe(5')) and BspHI (3'Probe) (NEB) were used in Southern blot (see Fig. 1F) The genomic DNA extracted from mouse tails after digestion was separated on a 1 % agarose gel and transferred to a positively charged nylon membrane (Hybond N+; Amersham International plc). The membrane was hybridized using a DIG Easy Hyb Granules (Roche Applied Science Inc.) at 42 °C overnight containing a PCR-generated probe, labeled by the PCR DIG probe synthesis kit (Roche Applied Science Inc.). Detection of hybridization signals was by using the DIG Luminescent Detection Kit (Roche Applied Science Inc.).

2.3. Animal breeding scheme

IL-1 β -IRES-Cre-ER^{T2} mice were generated via Crispr-Cas9 mediated recombination as described above. From this recombination, two IL-1 β -IRES-Cre-ER^{T2} founder mice (IL-1 β -TRAP) were generated. B6.Cg-Gt (ROSA)26Sortm9(CAG-tdTomato)Hze/J mice (ROSA26-LSL-tdTomato, JAX #007909) were purchased from Jackson Laboratories and after a 1 week of acclimation were mated with the IL-1 β -TRAP mice. Mice from the F0 generation of this cross that were heterozygous for both IL-1 β -TRAP and ROSA26-LSL-tdTomato were then crossed to

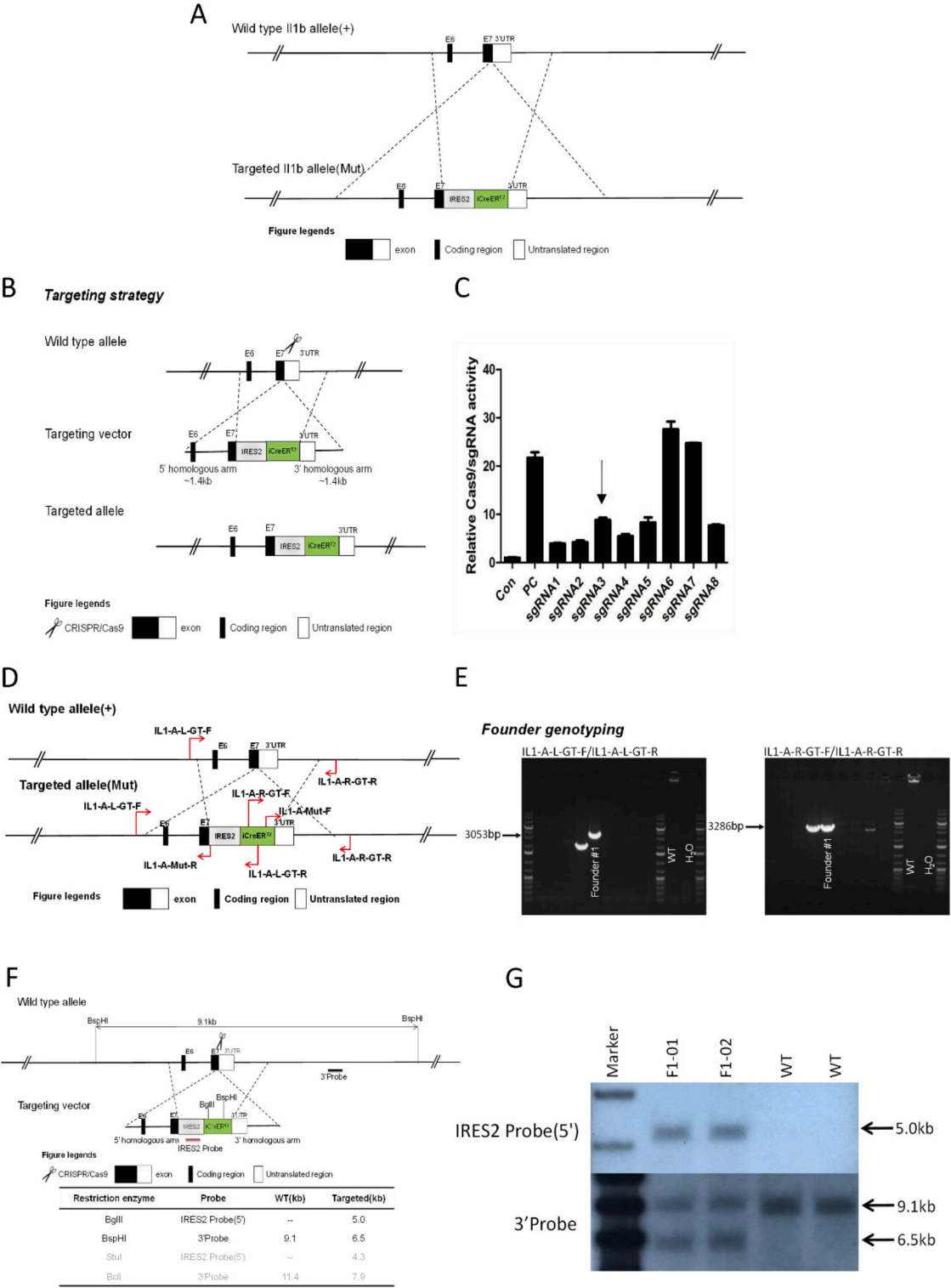


Fig. 1. Strategy and validation of creation of IL-1 β -TRAP founder mice. (A) Design schematic of Wild type and desired gene recombination following CRISPR/Cas9 mediated homologous recombination. Insertion of a IRES2-CreERT2 following the terminal stop codon in exon 7 of the Il1b allele prior to the 3' UTR. Black bars indicate coding regions exons and white bars indicate untranslated regions. (B) Design schematic of the targeting strategy. Scissor icon indicate the region where the WT Il1b allele was expected to be spliced to facilitate homologous recombination. The 5' and 3' homologous arms of the targeting vector were both ~ 1.4 kb and would result in the targeted allele. (C) Bar graph of relative Cas9/sgRNA activity based on the sgRNA amplicons used from [Supplemental Table 1](#). Arrow pointing to sgRNA3 was indicated as specific to the gene sequence while preventing non-specificity. (D) Design schematic of genotyping primers (red arrows) used to validate presence of the mutant gene at the junction site. (E) Photomicrograph of gel electrophoresis following genotyping founder mice with primer that detects the CreERT2 sequence within the 5' (left) or 3' (right) regions of the Il1b allele. Founder #1 was identified with both expected bands of 3053 bp (left) and 3286 bp (right) using primers as seen in [Supplemental Table 2](#). (F) Schematic design of southern blot of Il1b-IRES2CreERT2 to verify mutant gene is within the Il1b gene. The table below indicates the restriction enzymes, probes, and calculated amplicon length in WT and prospective Il1b-TRAP founder mice. (G) Photomicrograph of the southern blot of hybridized IRES Probe (5') and the 3' probe following incubation with restriction enzyme BspHI.

generate double homozygous IL-1 β -TRAP:ROSA26-LSL-tdTomato mice (IL-1 β -TRAP-reporter). Mice were group housed under standard 12 hr light/dark cycle conditions in rooms with controlled temperature and humidity. They were given standard rodent chow and sterilized tap water *ad libitum*. Genotyping was completed by Transnetyx. All experiments were approved by the Florida Atlantic University Institutional Animal Care and Use Committee.

2.4. Animal usage

12–18wk old male and female WT and IL-1 β -TRAP-reporter mice were used for all experiments. For baseline studies, tamoxifen was injected between 10–12 pm. For immunological stimulation, all tamoxifen was injected i.p. 12hrs prior to immune challenge of i.p. LPS, i.c.v. IL-1, or i.p. kainic acid (20 mg/kg). For kainic acid study, seizures were monitored for 2hrs following injection. All mice reached a seizure level of 3 or higher on the modified Racine scale. i.p. Diazepam (10 mg/kg) was used to stop convulsive seizures. Mice from all groups were euthanized by isoflurane overdose 7 days following tamoxifen injection for perfusion fixation.

2.5. Repetitive closed head injury (CHI)

Mice were group housed under standard 12 hr light/dark cycle conditions in rooms with controlled temperature and humidity. They were given standard rodent chow and sterilized tap water *ad libitum*. Genotyping was completed by Transnetyx. All experiments were approved by the University of Kentucky Institutional Animal Care and Use Committee and were carried out per the Guide for the Care and Use of Laboratory Animals and ARRIVE guidelines. IL-1 β -TRAP-reporter mice were 7.6 $\pm$ 1.5 months old (mean $\pm$ SD) at the time of surgery. Four groups were included: IL-1 β -TRAP-Cre negative littermates treated with tamoxifen (3F, 1 M), IL-1 β -TRAP-Cre positive mice not treated with tamoxifen (2F, 1 M), IL-1 β -TRAP-Cre positive mice treated with tamoxifen receiving either a sham (4F, 1 M), or CHI (3F, 2 M) surgery. Mice received two CHI or sham procedures separated by a 24-hour interval, following previously described methods (Bachstetter et al., Apr. 2020; Macheda et al., 2022). For surgery, mice were anesthetized with 4–3.5 % isoflurane in 100 % O₂ and maintained under anesthesia with continuous inhalation of isoflurane (2.0–3.0 %, 100–200 mL/min) in room air through a nose cone using the Kent Scientific SomnoSuite. Anesthesia depth was monitored by assessing the pedal withdrawal reflex. In the stereotaxic frame, a midline sagittal scalp incision exposed the coronal and sagittal sutures. A 1 ml latex pipette bulb filled with water was placed under the mouse's head. A single controlled midline impact was administered at 5.0 $\pm$ 0.2 m/s velocity, 100 ms dwell time, using a 5.0 mm flat steel tip, at coordinates relative to bregma: medio-lateral, 0.0 mm; antero-posterior, 1.6 mm; and 1.2 mm deep, using a Neuropactor (Neuroscience Tools). Sham-operated mice underwent the same procedure without impact. After 24 h, the surgery was repeated. No animals died or were excluded. The sham group's mean righting reflex times for surgeries 1 and 2 were 143.6 s (SD = 37.3) and 109.6 s (SD = 78.6), while the CHI group's times were 547.4 s (SD = 279.4) and 451.8 s (SD = 347.6), respectively. Mice received a single i.p. injection of 150 mg/kg tamoxifen (Sigma-Aldrich, CAS# 10540–29-1, Catalog no: T5648-5G) in corn oil 21 h after the first injury and 3 h before the second CHI. Seven days post-first injury, mice were anesthetized with 5 % isoflurane and transcardially perfused with ice cold 1X phosphate-buffered saline for 5 min. Brains were rapidly removed, fixed in 4 % PFA for 24 h, then transferred to 30 % sucrose. Skull caps with attached meninges were immersed in 4 % PFA for 6 h, then transferred to 30 % sucrose. Tissues were stored at 4 °C in the dark. An independent investigator randomized mice, balancing group representation in cages and ensuring equal male/female distribution. A computer-generated sequence randomized sham and CHI surgery order, with no more than two consecutive sham or CHI surgeries. The tamoxifen administrator

was blind to genotype and surgery status, while the surgeon was blind to genotype and tamoxifen conditions. Blinding was maintained by coding animals and treatments until data analysis completion.

2.6. Drug administration

All compounds were created fresh, aseptically, and used within 8–12hrs of their creation. Tamoxifen was prepared at a concentration of 20 or 40 mg/mL in sterile corn oil and heated for 37 °C for 8–12hrs to dissolve. Lipopolysaccharide (LPS, E.Coli 055:B5, Sigma) was prepared from a frozen stock solution of 2.5 mg/mL and diluted in sterile PBS (Gibco) to a concentration of 0.025 mg/mL. LPS was administered at 1 mg/kg for all experiments. Kainic acid was prepared immediately prior to injection and administered i.p. at a dose of 20 mg/kg in 100 μ L.

2.7. Stereotaxic surgeries

Stereotaxic surgery was performed according to Nemeth et. al (2024) (Nemeth et al., 2024). In brief, mice were deeply anesthetized with a 5 % isoflurane induction, followed by maintenance of anesthetic depth at 1–1.5 % isoflurane with a Somnosuite low-flow anesthesia system using room air (Kent Scientific). Body temperature monitoring & homeothermic control was automatically controlled by the Right Temp® heating pad and body temperature sensor (Kent Scientific). Upon induction, mice were placed onto the stereotaxic frame, secured to the nose cone, and head was secured with ear bars (David Kopf Instruments). For pre-operation pain management, all mice received 100 μ L of lidocaine/bupivacaine mixture directly under the expected incision site. The skin was disinfected with betadine and 70 % ethanol and a midline incision was cut to reveal the skull. Coordinates were precisely located by a semi-automated stereotaxic machine equipped with motorized directional arms (Neurostar GmbH). For lateral ventricle injections (AP, –0.6 mm; ML, –1.0 mm; DV, –2.5 mm), a 0.6 mm burr hole was automatically drilled and drilling was stopped using the AutoStop feature (Neurostar GmbH) to prevent additional brain damage. The automated nanoinjection robot affixed with a pulled glass capillary needle was then used to precisely locate the injection location and penetrated the brain at a rate of 1 mm/min. Once at the injection site, 2 μ L of sterile PBS or 20 ng of recombinant mouse Interleukin-1 β (R&D Systems) in 2 μ L of PBS was injected over 5mins which was controlled by the Stereodrive software (Neurostar GmbH). Surgical wounds were closed with Vetbond (3 M) and mice were allowed to fully recover in a heated recovery chamber. For post-operation pain management, all mice receive a single dose of buprenorphine-SR (0.36 mg/kg, s.c.) and were monitored daily for 5 days for any complications.

2.8. Immunohistochemistry

Mice were euthanized by isoflurane overdose and then transcardially perfused at a rate of 4 mL/min with a peristaltic pump with 16–20 mL of cold 1x PBS (pH 7.4, Gibco) followed by 24 mL of cold 4 % Paraformaldehyde (PFA, Sigma). Brains were removed and post-fixed in 4 % PFA overnight at 4 °C then equilibrated in 20 % sucrose containing 0.1 % sodium azide in PBS (pH 7.4) for 24hrs. 40 μ m sections were generated via cryostat and collected into 96-well plates filled with cryoprotectant solution (30 % ethylene glycol, 30 % polyethylene glycol in 0.1 M Phosphate buffer). Sections of the entire brain was collected in a sequential manner as described in Zhao and van Praag, 2020 (Zhao and van Praag, 2020). A single column of tissue in the 96-well plate was used for each study and this process was similarly employed in tissues collected in all animals. Sections were washed in three, 5-min washes of PBS, blocked with 5 % Normal Donkey Serum in 0.1 % Triton-X in PBS at RT for 1hr, then incubated overnight at 4 °C with primary antibodies: rabbit-anti-Iba-1 (WAKO, 1:1000), goat- anti-tdTomato (Origene Scigen, 1:500), rat-anti-CD45 (Biolegend, 1:2000), rat-anti-MHC Class II, I-A/I-E (Cell Signaling Tech, 86285S, 1:200), rabbit-anti-P2YR12 (Anaspec,

Cat#: AS-55042A, 1:2000), goat-anti-CD206 (R&D Systems, 0.2 mg/mL, AF2535, 1:1000). Sections were washed in PBS and then incubated with a fluorochrome conjugated secondary antibody (Alexa Fluor 488 nm, 594 nm, or 647 nm, ThermoFisher Scientific) for 2 hr at RT. Sections were then washed with PBS, plated on a microscope slide (Superfrost Plus, ThermoFisher) in 0.1 % Triton-X in PBS and cover slipped with Prolong Gold anti-fade mounting media containing DAPI (Invitrogen, CAT#P36971).

2.9. Whole mount choroid plexus

Mice were euthanized with overdose of isoflurane and perfused with 10 mL of cold 1x PBS (pH 7.4, Gibco). Brains were removed, bisected, the ventricles were opened, and the choroid plexus of the lateral ventricles were removed as previously described (Amaia et al., 2022). Tissue was fixed in 4 % PFA for 5 min then stored in 1x PBS at 4°C. Immunofluorescent labeling was conducted free-floating in 24 well plates similarly to brain tissue with Rat-anti-CD31 (MEC13.3, ThermoFisher, 1:200) and Goat-anti-tdTomato (Cat#: AB8181-200, Origene Sicgen, 1:1000). Sections were washed in PBS and then incubated with a fluorochrome conjugated secondary antibody (Alexa Fluor 488 nm, 594 nm, or 647 nm, ThermoFisher Scientific) for 2 hr at RT. Whole tissue were then washed with PBS, plated on a microscope slide (Superfrost Plus, ThermoFisher) in 0.1 % Triton-X in PBS and cover slipped with Prolong Gold anti-fade mounting media containing DAPI (Invitrogen, CAT#P36971).

2.10. For CHI experiments

Whole brains were coronally sectioned (30 µm thick) using a freezing microtome. Immunofluorescence staining was performed on free-floating sections and whole-mount meninges. Every 10th section between 1.3 mm and 2.5 mm relative to bregma was analyzed. Primary antibodies were: rabbit anti-GFAP (1:10,000, Dako, Z0334, AB_10013382) and guinea pig anti-Iba-1 (1:10,000, Synaptic Systems, HS-234, AB_2891296). AlexaFluor secondary antibodies were used. Primary antibodies were incubated for 12–16 h at 4 °C. TrueBlack treatment (Biotium, Cat number 23007) was used to block autofluorescence. TrueBlack solution (1x) was prepared by mixing 50 µL of 20x stock with 1 mL of 70 % ethanol. Sections were treated for at least 5 min, followed by three 5-minute water washes. Sections were cover slipped with Vectashield Vibrance H-1800 containing DAPI.

2.11. iDISCO+

Mice were euthanized by isoflurane overdose and then transcardially perfused as previously described. Brains were stored in 1xPBS containing 0.1 % Sodium Azide until the start of the iDISCO protocol. iDISCO was performed as previously described (Renier et al., 2014). On Day 1, the tissue was washed three times in PBS (30 min each) and dehydrated through a series of methanol steps (20 %, 40 %, 60 %, 80 %, 100 % MeOH, 1 h each) at room temperature. This was followed by overnight delipidation in 66 % DCM/33 % MeOH. On Day 2, dehydration continued with two 1-hour incubations in 100 % MeOH, followed by overnight bleaching in 5 % H₂O₂/MeOH at 4 °C. On Day 3, the tissue was rehydrated in a reverse methanol series (80 %, 60 %, 40 %, 20 % MeOH, 1 h each) and washed in 0.2 % Triton-X100 in PBS (PBST), followed by quenching in PBST with 20 % DMSO, and 0.3 M glycine for 36 h at 37 °C. On Day 5, blocking was performed in PBST with 10 % DMSO, 6 % NDS for 48 h at 37 °C. On Day 7, the tissue was incubated with the primary antibody (Rabbit-anti-tdTomato, 1:500, Rockland) in incubation solution (PBS with 0.2 % Tween20, 5 % DMSO, 3 % NDS, and 10 µg/mL heparin) for 7 days at 37 °C. On Day 14, extensive washing was performed over 12 h (with washes every 2 h) followed by an overnight wash in incubation solution at 37 °C. On Day 15, the tissue was incubated with the secondary antibody (Donkey-anti-rabbit Alexa Fluor 647,

1:500) under the same conditions for another 7 days at 37 °C. On Day 22, the tissue underwent additional washing. On day 23, tissues underwent dehydration through a methanol series, delipidation, and final clearing in dibenzyl ether (DBE) until transparency was achieved. Whole brains were mounted to an Ultramicroscope Blaze LSMF and scanned with the 4x objective with 1.5x zoom on both 488 nm and 647 nm channels. All stitching, 3D reconstructions, and.mp4 visualizations were completed in Imaris (Oxford Instruments, v10.1). The 488 nm channel was used as a background channel to capture autofluorescence signals.

2.12. Imaging and image analysis

Brain sections were selected for imaging between AP:-0.2 mm and -0.8 mm relative to bregma which contains all the brain regions for further analysis: cortex (CTX), hypothalamus (HY), choroid plexus (ChPlx), subfornical organ (SFO), and striatum (STR). Two images per region per animal were taken for quantification. Representative images and images for analysis were taken with Nikon A1R confocal microscope as multi-channel Z-stacks using the excitation wavelengths of 405 nm, 488 nm, 564 nm, and 670 nm. Multi-channel Z-stacks were then projected using the Max Intensity Projection (MIP) plugin with FIJI (ImageJ 2.0, Bethesda, NIH). Multi-channel MIP images were loaded into open-source, pixel classification machine learning software Illastik (Berg et al., 2019). 3 training images per treatment were used to train the pixel classification. Following training, Illastik was used to produce binary masks of each channel. The binary masks, per channel, were applied onto the original images as background subtraction. Images were then quantified using stereological methodology. Using a custom FIJI macro, images were divided into a grid containing 20 equally sized images and 5 of those grid boxes were randomly selected for quantification. A trained, blinded observer then counted the single positive, double positive, and triple positive cells per field. Cell counts were estimated by dividing number of cells by ROI area sampled. For whole mount choroid plexus tissue, tile scan, z-stacked images were acquired to visualize the entire tissue and stitched with Nikon NIS Elements software.

2.13. For CHI experiments

Images were captured using a Nikon A1R confocal microscope (40x objective, 0.5 µm z-step) and viewed with Imaris 10.1. Widefield epifluorescence imaging of whole brain slices was performed using a Zeiss AxioScan Z.1 (20x objective, 1 µm z-step). Images were projected into 2D using maximum intensity projection and viewed with HALO software (Indica Labs Inc., Albuquerque, NM, USA). tdTomato⁺ cells were counted over three brain sections per animal for both hemispheres, adjusted for total tissue volume. Only objects resembling cells (i.e., not specks) were counted. The choroid plexus was excluded from all samples as it was not uniformly present in the tissue, and many tdTomato-positive cells were found there. Cell counting was performed blind to experimental conditions.

2.14. RNA isolation and qRT-PCR

Brains from 14wk old mice (n = 7) were collected and the hippocampus and choroid plexus were isolated and collected into sterile Eppendorf tubes. The 4th ventricle and lateral ventricle choroid plexus were dissected using fine tip forceps and pooled into one tube per animal. For hippocampal RNA isolation, the tissue was suspended in Tri-Reagent (Millipore-Sigma, Cat#:93289), homogenized with a sonicator, and total RNA was isolated according to manufacturer's instructions. For the choroid plexus, the RNeasy Micro kit (QIAGEN, Cat#: 74004) was used to isolate RNA as detailed by the manufacturer. RNA concentration and purity were determined with a spectrophotometer. RNAs were reverse transcribed into cDNAs using the High-Capacity cDNA Reverse Transcription Kit (Applied Biosystems, Cat#: 4368814). RT-qPCR analysis was performed for *Il1b* gene expression using Taqman

Universal Mastermix (ThermoFisher, Cat#:4304437) and the following primers: *Il1b* (Mm01336189_m1, FAM) and primer-limited GAPDH (Mm99999915_g1, VIC) primers. Gene amplification curves were measured on an ABI PRISM 7500 sequence detection system (Applied Biosystems, Foster, CA) by real-time PCR. *Il1b* gene expression was normalized based on reference cDNA levels of GAPDH. Relative gene expression was analyzed using the $\Delta\Delta CT$ method and results are expressed as paired analysis of the fold change from each individual animal's hippocampus.

2.15. Statistical analysis

All data is expressed as treatment mean $\pm$ standard deviation of the mean. Direct comparisons of normally distributed data were analyzed with two-tailed Student's T-Test with alpha set to $p < 0.05$ as the criterion for the rejection of the null hypothesis. Paired analyses of non-normal data were analyzed with a two-tailed Wilcoxon matched-pairs signed rank test with alpha set to $p < 0.05$ as the criterion for the rejection of the null hypothesis. Two-way ANOVA was used for comparison of multiple groups. When appropriate, significant main and interaction effects were subjected to Bonferroni test for post hoc analyses for further familywise comparison. All data were analyzed using GraphPad Prism version 8.0.1 for Windows, (GraphPad Software, Boston, Massachusetts USA, <https://www.graphpad.com>).

2.16. Contact for reagents and data sharing

Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Ning Quan (nquan@health.fau.edu).

3. Results

3.1. Generation of the IL-1 β -TRAP mouse model

In this study, a novel knock-in mouse line to detect IL-1 β -expressing cells was created by precisely inserting an IRES-CreER^{T2} sequence in the *Il1b* gene after the stop codon in exon 7 before to the 3' untranslated region (Fig. 1A). To enact the targeting strategy (Fig. 1B), sequences were selected which span the exon 7 and 3'UTR. These sequences were synthesized into single guide RNA (sgRNA) constructs (Supplementary Table 1). 8 of the sgRNAs were then tested for off-target cleaving capacity using Universal CRISPR Activity Assay (Biocytogen). Compared to positive controls, sgRNA6 and 7 were detected to have off-target effects and sgRNA3 was determined to have elevated efficacy without off-target activity and, was determined to be appropriate for use (Fig. 1C). Confirmation of Cas9/sgRNA3 editing was carried out by gel electrophoresis of PCR products in WT and mutant mice which identified Founder #1 (Fig. 1D-E). Sequences of primers used are listed in Supplementary Table 2. To determine heritability, this founder was crossed with a.

WT C57/BL6 mouse and offspring were genotyped. Isolated DNA was also subjected to Southern blot to verify that the location of insertion was within the correct gene locus and no off-target effects occurred. Using the probes and restriction enzymes from Supplementary Fig. 1 (design of the CRISPR knock-in strategy), restriction mapping identified F1-01 and F1-02 as the first IL-1 β -TRAP mice. F1-01 and F1-02 were bred with ROSA26-LSL-tdTomato mice to generate IL-1 β -TRAP-reporter mice which are used in the rest of the study.

3.2. The choroid plexus is the location of basal IL-1 β production in the brain

To determine where IL-1 β is expressed in the normal brain, IL-1 β -TRAP-reporter mice were injected once with tamoxifen (150 mg/kg) or corn oil and brains were isolated 7d later for IHC or lightsheet

microscopy (Fig. 2A), as previously described (Guenther et al., 2013). Full brain imaging using light sheet microscopy of an IL-1 β -TRAP-reporter brain ($n = 1$) revealed that tdTomato expression was extensive throughout the ventricles; however, scattered sparsely in the circumventricular organs, such as the median eminence (ME) and subfornical organ (SFO), and throughout the meninges (Fig. 2B). Sporadic tdTomato expression was found in vasculature and, within the brain parenchyma, very few, if any, cells were labeled (Supplementary Video 1). Further, in brain sections, triple immunofluorescent labeling of tdTomato, Iba-1, and CD45 was conducted and the choroid plexus (ChPlx), subfornical organ (SFO), cortex (CTX), striatum (STR) and hypothalamus (HY) were imaged (as indicated in Fig. 2C). In control corn oil injected IL-1 β -TRAP-reporter mice ($n = 3$), tdTomato was not detected throughout the brain parenchyma or ChPlx (Supplementary Fig. 1A,B). However, a dim tdTomato signal was detected in corn oil injected IL-1 β -TRAP-reporter mice and tamoxifen injected IL-1 β -TRAP Cre-negative mice (Supplementary Fig. 1C-D), suggesting the ROSA26-LSL-TdTomato may have a low above background signal that should be discounted as artifact, which has been documented before (007914 - Ai14, Ai14D or Ai14 (RCL-tdT)-D Strain Details). Throughout the brain parenchyma (STR, HY, CTX) of TAM injected mice, out of 492 Iba-1⁺ cells counted, only 2 cells were identified to be tdTomato positive (Fig. 2E and Supplementary Fig. 1E-F). No tdTomato⁺ cells were found in the SFO (Supplementary Fig. 1E). In the ChPlx of tamoxifen injected mice, out of all tdTomato cells counted ($n = 67$ cells from 4 mice), 6.8 % $\pm$ 13.6 were CD45⁺, 7.5 % $\pm$ 9.57 were Iba-1⁺, and the majority, 85.68 % $\pm$ 11.89, were identified as Iba-1⁺CD45⁺ cells (Fig. 1D and Supplementary Fig. 1E1).

To assess IL-1 β -expression levels and patterns in the choroid plexus, whole choroid plexus tissue from the lateral ventricles were collected for *Il1b* mRNA expression, tdTomato-expressing cell density, and various macrophage cell surface markers. *Il1b* gene expression was measured by qPCR in microdissected choroid plexus (ChPlx) and hippocampus (HC) tissue. A Wilcoxon signed-rank paired analysis of HC vs ChPlx from WT mice ($n = 7$) showed that there is significantly more *Il1b* mRNA in the ChPlx (between 10–22-fold higher expression) than the HC ($p = 0.0156$, Sum of signed ranks (W) = 28.00, 95 %CI = 9.701 to 25.32) (Fig. 2F). Wholemount choroid plexus from the lateral ventricles (LV-ChPlx) of corn oil or tamoxifen injected mice were collected and immunolabeled for tdTomato. LV-ChPlx from tamoxifen treated mice (TAM) had a statistically significant increase in density of tdTomato-expressing cells, approximately 500 $\pm$ 50.32 tdTomato⁺ cells/mm² in TAM injected mice compared to 60 $\pm$ 56.55 cells/mm² in corn oil injected mice (Fig. 2G-H, $t = 9.321$, $df = 3$, $p = 0.0026$). We hypothesized that the tdTomato-expressing cells that were found in the choroid plexus tissue sections were macrophages, as they were identified as Iba-1 and CD45 positive in the tissue sections. It is well established that choroid plexus and border associated macrophages (BAM) express various cell surface markers such as MHCII and CD206. LV-ChPlx whole mounts immunolabeled for DAPI, MHCII, tdTomato, and CD206 show, compared to corn oil controls, the IL-1 β -expressing cells are primarily MHCII expressing or CD206/MHCII double positive macrophages (Fig. 2I). Two-way ANOVA revealed a significant interaction between macrophage population markers and treatment, $F(3, 12) = 4.81$, $p = 0.020$, indicating that treatment effects differed by macrophage population markers. There was also a significant main effect of treatment, $F(1, 12) = 18.78$, $p = 0.001$, but not of population, $F(3, 12) = 3.13$, $p = 0.066$. Bonferroni's multiple comparisons test showed that, relative to corn oil, the TAM treatment significantly increased the proportion of tdTomato⁺MHCII⁺ cells (95 % CI [180.3, 39.65], $p = 0.0025$) and tdTomato⁺CD206⁺MHCII⁺ cells (95 % CI [151.4, 10.77], $p = 0.0217$). No significant differences were observed for tdTomato⁺ or tdTomato⁺CD206⁺ populations (both $p > 0.9999$) (Fig. 2J, Supplementary Fig. 1H-K). MHCII and CD206 are not sufficient to determine whether the macrophages are stromal, epiplexus, or perivascular macrophages; therefore, we conducted immunofluorescent labeling for

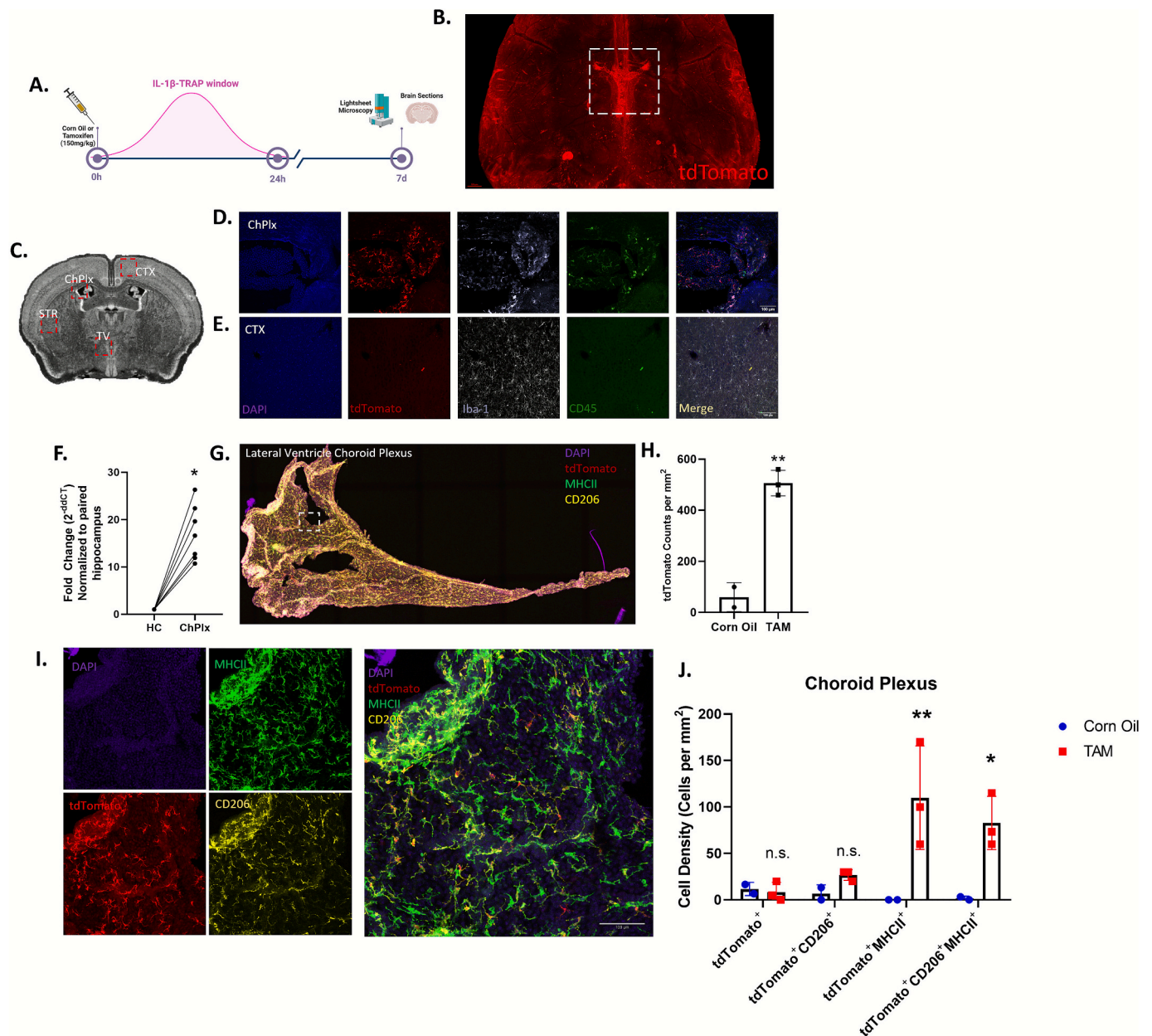


Fig. 2. Basal brain IL-1 β is exclusively expressed in choroid plexus macrophages. (A) Experimental design schematic for the study of basal *Il1b* expression. IL-1 β -IRES-Cre-ER^{T2}:ROSA26-lox-stop-lox-tdTomato mice were injected with corn oil or 150 mg/kg tamoxifen in corn oil. Pink line indicates expected duration of active recombination following a single i.p. injection of tamoxifen. 7d following i.p. tamoxifen mice were euthanized for light sheet microscopy or immunohistochemistry. (B) Light sheet microscopy maximum intensity projection image of tdTomato from the dorsal view of the IL-1 β -TRAP mouse brain given a single i.p. Tamoxifen (150 mg/kg). Dotted lines indicate the ventricles. (C) A representative nissl-stained coronal brain section highlighting the choroid plexus (ChPlx), cortex (CTX), striatum (STR), and hypothalamus (HY) with red dashed boxes for anatomical localization that were selected for imaging. (D-E) Immunofluorescent micrographs of DAPI (blue), tdTomato (red), Iba-1 (grey), CD45 (green) and merged channels of the lateral ventricle ChPlx (D) or the CTX (E) 7d following i.p. Tamoxifen (150 mg/kg). (F) Analysis of qRT-PCR of *Il1b* gene expression between the hippocampus (HC) and ChPlx of individual WT mice (n = 7). Each paired point represents matched tissue samples from the same mouse, with values normalized to the hippocampus *Il1b* ΔCT (dCT). Each black circle represents a sample, and the solid line indicates a paired sample of one individual. Wilcoxon matched-pairs signed rank test was conducted and data was presented as fold change respective to the individual mouse hippocampus. (G) Immunofluorescent tile scan of whole mount choroid plexus labeled for DAPI (purple), tdTomato (red), MHCII (green) and CD206 (yellow). Dotted white box indicates region shown in Fig. 2I. (H) Quantification of cell density (cells/mm²) of tdTomato-expressing cells in whole mount ChPlx. (I) 20x immunofluorescent images of DAPI (purple), tdTomato (red), MHCII (green), CD206 (yellow) and merged channels (right). Quantification of cell density (cells/mm²) of tdTomato expressing cells expressing cell surface markers of CD206, MHCII, or both. 2-way ANOVA was conducted. Data in H and J was presented as mean /pm SD. *p ≤ 0.05; **p ≤ 0.01 Fig. 2A was created in BioRender. Nemeth, D. (2025) <https://BioRender.com/m28r705>.

CD31 on the LV-ChPlx to identify relative locations of tdTomato expressing cells. No tdTomato-expressing cells were found attached to the CSF-facing, apical, choroid plexus epithelium, indicating that the tdTomato⁺ cells are likely not epiplexus macrophages. Additionally, tdTomato⁺ cells reside adjacent to or in between CD31⁺ vasculature,

which suggests IL-1 β expressing cells are either stromal or perivascular macrophages (Supplementary Fig. 1G). These data suggest, that under normal conditions, choroid plexus MHCII⁺ macrophages are the primary cell type within the brain with significant basal IL-1 β production.

3.3. Peripheral or central stimulus induced IL-1 β -expression

IL-1 β expressing cells were visualized after peripheral LPS-induced neuroinflammation. IL-1 β -TRAP-reporter mice were given an i.p. injection of tamoxifen 12hr prior to an i.p. injection of 1 mg/kg LPS or PBS. TdTomato was allowed to express for 7d and brains were taken for whole brain lightsheet imaging of tdTomato (n = 1) and standard immunofluorescent labeling (Fig. 3A). Lightsheet microscopy of LPS-treated IL-1 β -TRAP-reporter mice revealed a robust population of tdTomato⁺ cells at the brain's borders (Fig. 3B, Supplementary Video 2). Many of the tdTomato⁺ cells, resided near the leptomeninges, lined the vasculature, or occupied the choroid plexus. Within the brain parenchyma, many tdTomato⁺ cells with microglia morphology were observed throughout the entire brain. Compared to i.p. injections of PBS (n = 9), i.p. LPS (n = 5) groups showed an increase in tdTomato cells throughout CTX (Fig. 3D-E), STR (Supplementary Fig. 2A-B), and HY (Supplementary Fig. 2C-D). Within the STR, CTX, and HY cells that express tdTomato were observed to be greater than 90 % Iba-1⁺/CD45⁺ (CTX:85.41 % $\pm$ 19.96 (n = 77 cells); STR: 99.6 % $\pm$ 19.96 (n = 69 cells); TV: 91.66 % $\pm$ 11.91 (n = 69 cells) whereas the remainder were detected as CD45⁺ (Supplementary Fig. 1E). TdTomato⁺Iba-1⁺ cell density was assessed as a percentage of total Iba-1⁺ cells. In the CTX and STR, tdTomato expressing cells were 10.22 % and 11.84 % of the total Iba-1 population; whereas in HY, tdTomato-expressing cells contributed to 20.41 % of the total Iba-1⁺ population. Two-way ANOVA was performed to assess whether treatment and region of the brain influences tdTomato expressing microglia. The results indicated a significant main effect of treatment (F(1, 36) = 98.28P < 0.0001) and region (F(2, 36) = 9.237P = 0.0006) without interaction between treatment and region (F(2, 36) = 3.034P = 0.0606). Posthoc testing using Bonferroni multiple comparisons indicate LPS treatment increases tdTomato⁺Iba-1⁺ populations in the CTX (p = 0.0015), STR (p = 0.0002), and HY (<0.0001) compared to PBS controls. Among regions, the HY displayed the highest tdTomato

expression, with significantly greater levels than both the CTX (p = 0.0057) and STR (p = 0.0331) in LPS-treated mice. No significant differences were detected among PBS-treated groups across regions (all p > 0.99). Collectively, these data indicate that systemic LPS administration robustly increases tdTomato expression across brain regions, with the strongest effects observed in the hypothalamus (Fig. 3F).

Next, CNS-restricted neuroinflammation was elicited by direct i.c.v. injection of IL-1 β (Liu et al., 2019; Nemeth et al., 2024). IL-1 β -TRAP-reporter mice were given an i.p. injection of tamoxifen 12hr prior to an i.c.v. injection of IL-1 β (n = 4) or PBS (n = 3) (Fig. 4A). Compared to i.c.v. PBS controls, i.c.v. IL-1 β induced a marked increase in the number of tdTomato⁺ cells in the CTX (Fig. 4C-D), STR (Supplementary Fig. 3A-B), and HY (Supplementary Fig. 3C-D). TdTomato⁺Iba-1⁺ cell density was assessed as a percentage of total Iba-1⁺ cells. In the CTX and STR, tdTomato expressing cells were 6.9 % and 32.34 % of the total Iba-1 population, respectively and in the HY, tdTomato-expressing cells contributed to 17.41 % of the total Iba-1⁺ population. Two-way ANOVA was performed to assess whether treatment and region of the brain influences tdTomato expressing microglia. The results indicated a significant main effect of treatment (F(1, 15) = 10.87P = 0.0049) but not region (F(2, 15) = 0.9141P = 0.4221) or an interaction between treatment and region (F(2, 15) = 1.359P = 0.2868). Posthoc testing using Bonferroni multiple comparisons indicate i.c.v. IL-1 β treatment increases tdTomato⁺Iba-1⁺ populations in the STR (p = 0.0205) but not the CTX (p = >0.9999) or HY (p = 0.2944) compared to PBS controls. These data indicate that central IL-1 β administration modestly increases tdTomato expression in the HY, with the strongest effects observed in the STR (Fig. 4F).

I.c.v. IL-1 β administration causes peripheral leukocytes to enter the CNS (Liu et al., 2019; Proescholdt et al., 2002). Therefore, it is possible that peripheral leukocytes could be recruited to the brain, remain, and express Iba-1, similarly to microglia. To determine whether the tdTomato⁺Iba-1⁺ cells were indeed microglia, we co-labeled brain

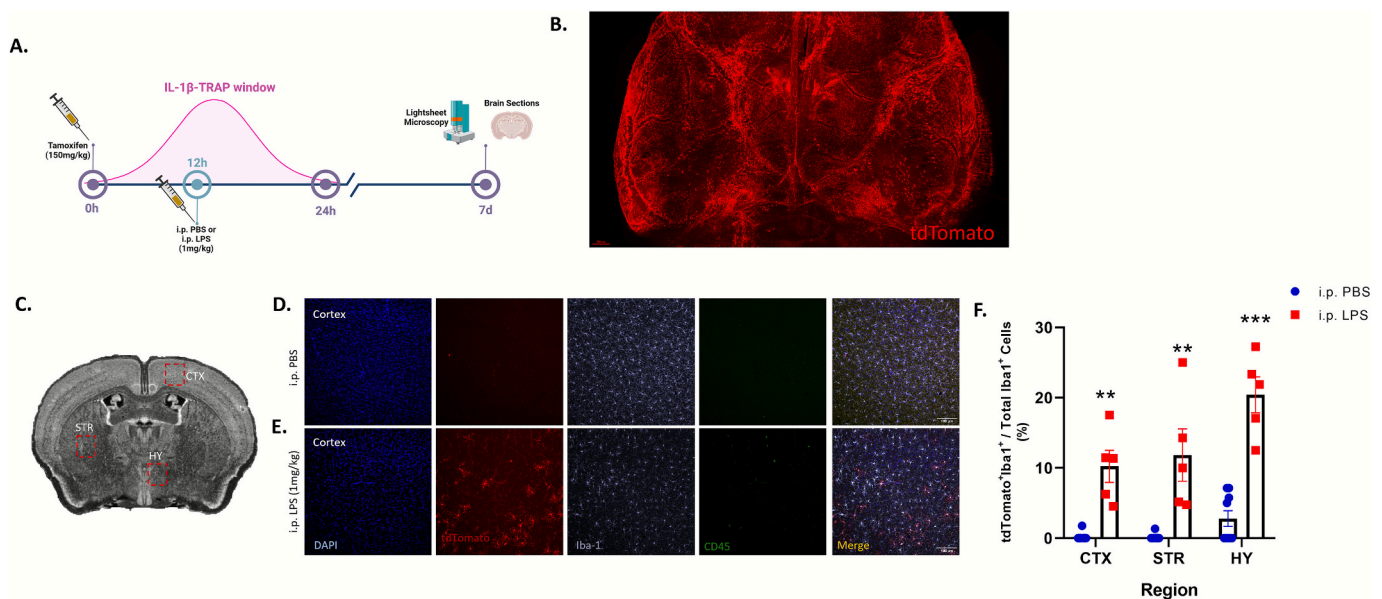


Fig. 3. Peripheral immune stimulus increases tdTomato expression at the brain's borders and within the brain parenchyma. (A) Experimental design schematic for the study of IL1 β expression following peripheral inflammation. IL-1 β -TRAP:R26-LSL-TdTomato mice were injected with 150 mg/kg Tamoxifen in corn oil and 12hr later were injected with i.p. PBS or i.p. LPS (1 mg/kg). Pink line indicates expected duration of active recombination following a single i.p. injection of tamoxifen. 7d following i.p. Tamoxifen mice were euthanized for light sheet microscopy or immunohistochemistry. (B) Maximum intensity projection from whole brain light sheet microscopy of tdTomato expression in a dorsal view of the brain from an IL-1 β -TRAP mouse treated with tamoxifen (150 mg/kg, i.p.) and LPS (1 mg/kg, i.p.). Scale bar = 500 μ m. (C) A representative nissl-stained coronal brain section highlighting cortex (CTX) striatum (STR), and hypothalamus (HY) with red dashed boxes for anatomical localization that were selected for quantification. (D-G) Immunofluorescent micrographs displaying DAPI (blue), tdTomato (red), Iba-1 (grey), CD45 (green), and merged channels in the CTX following i.p. PBS (D) or LPS (E). (F) Bar graph showing the percentage of TdTomato⁺Iba-1⁺ cells relative to total Iba-1⁺ cells following i.p. PBS (blue circles) and i.p. LPS (red squares) treatment in the cortex (CTX), striatum (STR) and hypothalamus (HY). Fig. 3A was created in BioRender. Nemeth, D. (2025) <https://BioRender.com/m28r705>.

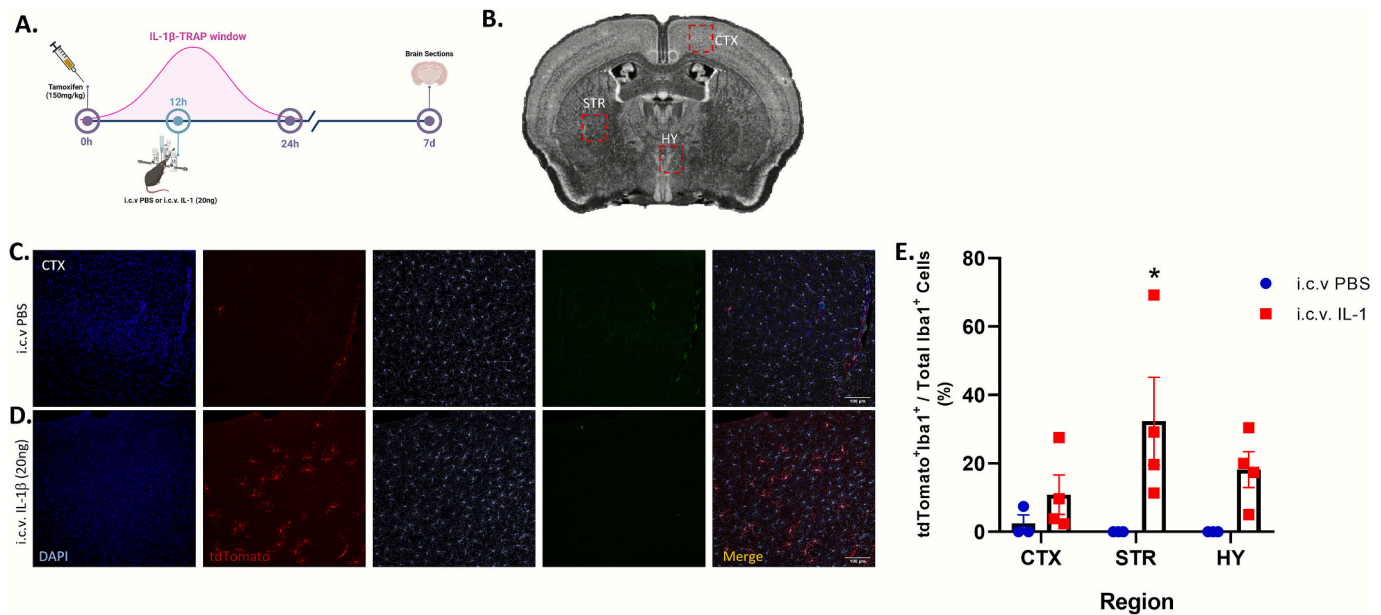


Fig. 4. Central immune stimulus increases tdTomato expression within the brain parenchyma. (A) Experimental design schematic for the study of IL1b expression following central inflammation. IL-1 β -TRAP-reporter mice were injected with 150 mg/kg Tamoxifen in corn oil and 12hr later were injected with i.c.v. PBS or i.c.v. IL-1 β (20 ng). Pink line indicates expected duration of active recombination following a single i.p. injection of tamoxifen. 7d following i.p. Tamoxifen mice were euthanized for immunohistochemistry. (B) A representative nissl-stained coronal brain section highlighting the subfornical organ (SFO), choroid plexus (ChPlx), and cortex (CTX) with red dashed boxes for anatomical localization that were selected for imaging. (B-G) Immunofluorescent micrographs showing DAPI (blue), tdTomato (red), Iba-1 (grey), CD45 (green), and merged channels in the (B-C) choroid plexus (ChPlx), (D-E) subfornical organ (SFO), and (F-G) cortex (CTX) 7 days after i.c.v. injection of PBS (B, D, F) or IL-1 β (C, E, G). Scale bar = 100 μ m. (H) Bar graph showing the percentage of TdTomato⁺ Iba1⁺ cells relative to total Iba1⁺ cells following i.c.v. PBS (blue circles) and i.c.v. IL-1 β (red squares) treatment in the cortex (CTX), striatum (STR) and hypothalamus (HY). Fig. 4A was created in BioRender. Nemeth, D. (2025) <https://BioRender.com/m28r705>.

sections from i.c.v IL-1 β injected mice with Iba-1, tdTomato and P2YR12 (Supplementary Fig. 3E). In i.c.v IL-1 β injected mice, 96.33 % $\pm$ 2.510 of the tdTomato⁺ cells were also double positive for Iba-1 and P2YR12 ($t = 36.92$, $df = 3$, $p = <0.0001$). Only 3.675 % $\pm$ 2.510 were tdTomato⁺ but neither Iba-1 or P2YR12 positive (Supplementary Fig. 3F).

3.4. Novel neuronal IL-1 β -expressing cells under various conditions

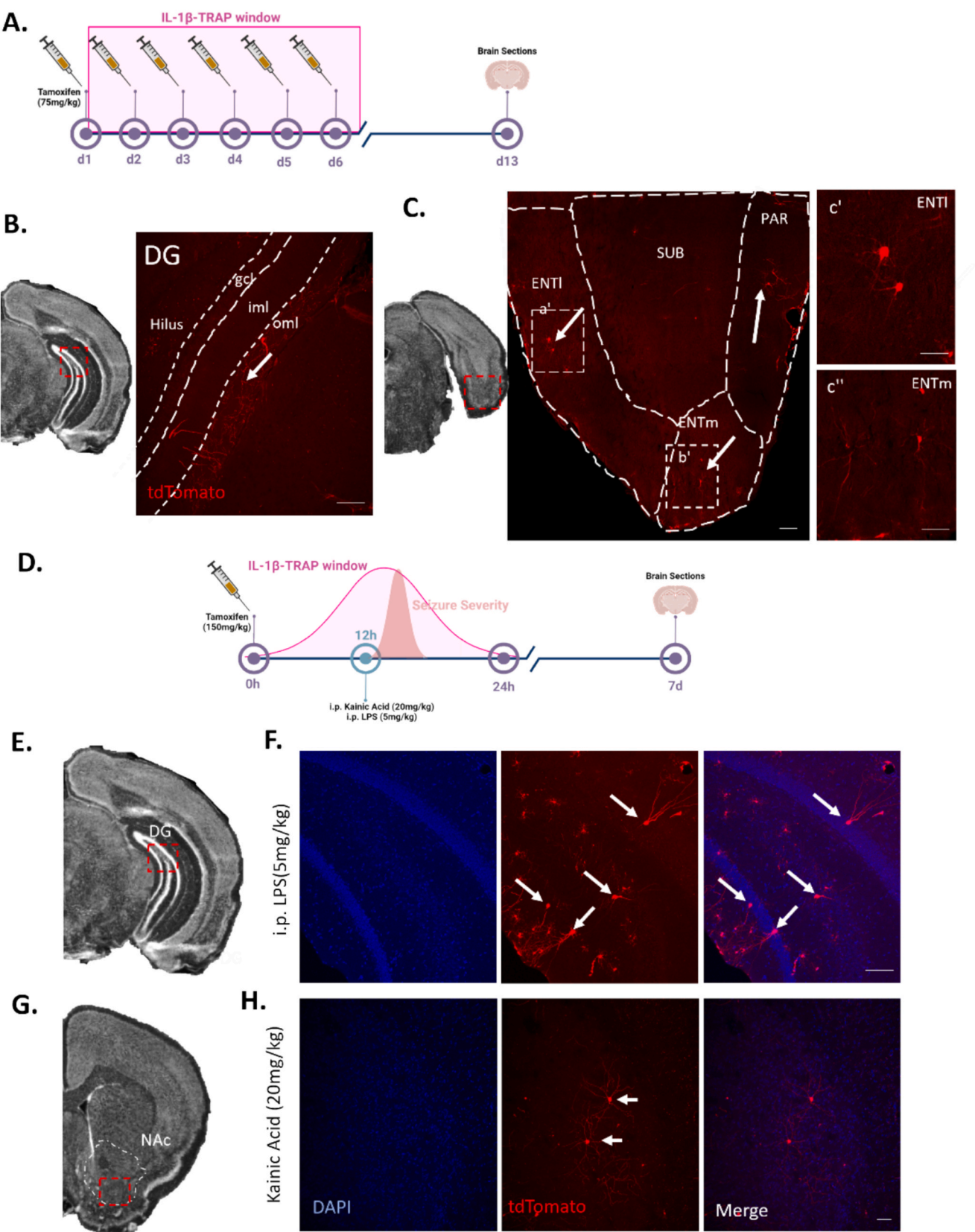
IL-1 β -TRAP-reporter mice were subjected to 6 consecutive days of tamoxifen injection (6xTAM, 75 mg/kg) to track the IL-1 β expression across the course of a week (Fig. 5A). Following 6xTAM, choroid plexus macrophages were still present and 6xTAM mice had noticeably more tdTomato-expressing microglia throughout the entire brain parenchyma. These microglia were stochastically distributed throughout the entire brain (Supplementary Fig. 4A-B). In the 6xTAM mice, presence of fibrous staining in the dentate gyrus outer molecular layer was detected (Fig. 5B). The Allen Brain Mouse Brain Connectivity atlas was used to determine the possible source of the tdTomato⁺ cell fibers. From this search, the medial and lateral entorhinal cortex (ENT) were then imaged (Fig. 5C). TdTomato⁺ expressing neurons were found in the lateral ENT, medial ENT and parasubiculum (PAR). These tdTomato⁺ neurons were always found in pairs and distributed sporadically in these regions. This observation leads us to believe that there are long-projecting neurons from the ENT and PAR that express *Il1b* mRNA. These projections may innervate neuronal IL-1R1-expressing regions like the dentate gyrus; however, whether these neurons actually secrete bio-active IL-1 β is unknown.

Cytotoxic and highly neuroinflammatory stimuli were also tested to determine if magnitude or type of neuroinflammation can change the expression patterns of *Il1b*. In the first experiment, IL-1 β -TRAP-reporter mice were given a large, near-septic, dose of LPS (5 mg/kg) 12hr after

administration of tamoxifen ($n = 5$). 7d later mice were sacrificed and brain tissues were imaged for tdTomato expression. There were many tdTomato-expressing cells in the brain parenchyma and most of them were microglia; however, in the hippocampus neurons expressing tdTomato were found. These tdTomato⁺ neurons were scattered throughout the granule cell layer of the dentate gyrus and in CA4 (Fig. 5E-F). In the second experiment, 12hr following tamoxifen, IL-1 β -TRAP-reporter mice were given i.p. kainic acid (20 mg/kg) and seizures were monitored for 2hrs. All mice reached a seizure level of 3 or higher on the modified Racine scale. Diazepam (10 mg/kg) was used to stop convulsive seizures. No induction of microglia IL-1 β was observed. However, the only induced IL-1 β following kainic acid were nucleus accumbens medium spiny neurons (Fig. 5G-H). Taken together these data reveal that specific neuroinflammatory conditions can induce the expression of IL-1 β mRNA in unique neuronal populations.

3.5. IL-1 β distribution following TBI

It is also well known that traumatic brain injury is associated with increased cytokine production, including IL-1 β , but the cellular source of IL-1 β is controversial (Vincent et al., 2023). Some studies report that IL-1 β is primarily produced by microglia after injury, while others suggest that endothelial cells are the main producers (Chung et al., 2019). Following a mild TBI in mice caused by a closed head injury (CHI), the peak of injury-induced IL-1 β -expression occurs 9 h after the injury. Therefore, IL-1 β -TRAP-reporter mice were used to identify the IL-1 β -producing cells during this timeframe post CHI. To test this, a two-hit model of repetitive TBI (Bachstetter et al., Apr. 2020) was used and single injection of Tamoxifen (150 mg/kg i.p.) was administered 3 h before the second CHI surgery (Fig. 6A). We estimated that this procedure would label the peak of *Il1b* gene expression and efficiently tag cells producing IL-1 β during this period (Guenther et al., 2013). 7 days after the injury, IL-1 β -producing cells and their proximity to other



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Fig. 5. Neuronal *Il1b* expressing cells in homeostasis or following immune or excitatory challenges. (A) Experimental design schematic for the study of *Il1b* expression across multiple days. Pink box indicates expected duration of active recombination following a single i.p. injection of tamoxifen. IL-1 β -TRAP-reporter mice were injected with 75 mg/kg Tamoxifen in corn oil daily for 6d. Mice were euthanized for immunohistochemistry 7d following the final tamoxifen injection. (B) Immunofluorescent micrographs of tdTomato expression in the outer molecular layer (oml) of the dentate gyrus and (C) lateral (lENT)/ medial (mENT) entorhinal cortex. Dashed lines indicate regional separation between features. gcl = granule cell layer; iml = inner molecular layer; oml = outer molecular layer; ENTl = lateral entorhinal cortex; ENTm = medial entorhinal cortex; PAR = parasubiculum; SUB = subiculum. Scale bar = 100 μ m. (D) Experimental design schematic for the study of *Il1b* expression following a large inflammatory or excitotoxic insult. Pink line indicates expected duration of active recombination following a single i.p. injection of tamoxifen. Pink line indicated the duration and intensity of seizures following kainic acid. IL-1 β -TRAP-reporter mice were administered a single injection of tamoxifen (150 mg/kg) 12hrs prior to either i.p. LPS (5 mg/kg, $n = 4$) or i.p. KA (20 mg/kg, $n = 3$) and were euthanized 7d following injection. (E) Representative nissl-stained sections indicating area of imaging of the DG following 5 mg/kg i.p. LPS. (F) Immunofluorescent images of DAPI, tdTomato, and merged channels show tdTomato⁺ neurons in the dentate gyrus in the hippocampus following the LPS challenge ($n = 4$, F). (G) Representative nissl-stained sections indicating area of imaging of the nucleus accumbens (NAc, white dotted line) 7d following i.p. kainic acid. (H) Immunofluorescent images of DAPI, tdTomato, and merged channels show tdTomato⁺ neurons following kainic acid administration. White arrows in F and H indicate morphologically identified neurons. Scale bar = 100 μ m. Fig. 5A and D were created in BioRender. Nemeth, D. (2025) <https://BioRender.com/m28r705>.

pathological neuroinflammatory changes were examined.

In addition to collecting the brain, dural meningeal wholemounts were collected, as the above results have highlighted the significant role of the meninges in immune signaling (Fig. 6B). In sham-injured animals, tdTomato-positive cells had little overlap with either Iba-1 or GFAP markers (Fig. 6C). After a CHI, there was an increase tdTomato positive cells, and most of these cells overlapped with Iba-1. To determine whether tdTomato expression marked a morphologically distinct subset of microglia after controlled head injury (CHI), we quantified Iba-1⁺ cells across entire coronal sections from 3 to 5 regions per animal using HALO AI. On average, 3,000–7,000 microglia were identified per mouse. tdTomato⁺ cells were rarely observed in sham animals and therefore analyses focused on CHI brains. Across CHI animals, 14,826 tdTomato[−] and 58 tdTomato⁺ microglia were identified. The mean Iba-1⁺ cell area was not significantly different between tdTomato[−] and tdTomato⁺ populations (161.6 ± 1.2 vs. $158.7 \pm 19.8 \mu\text{m}^2$; unpaired t test, $t(14,882) = 0.15$, $p = 0.88$, Supplementary Fig. 5E). At the animal level, tdTomato⁺ cells represented a small fraction of the total microglial population ($0.43 \pm 0.11\%$, $n = 3$ animals; Supplementary Fig. 5F). Thus, tdTomato⁺ microglia are present following CHI but constitute a rare population without distinct morphological differences compared to the broader Iba-1⁺ pool. In the brain, sparse labeling of tdTomato⁺ cells in the sham-injured animals, with no consistent pattern of their location was observed (Fig. 5E–G), except for the choroid plexus, which showed robust tdTomato positive cells as noted earlier (Supplementary Fig. 5A). In CHI-injured mice, we found a greater number of tdTomato⁺ cells throughout the brain parenchyma. The spatial distribution of tdTomato-positive cells after injury showed a pattern of cells in the cortex directly under the impact device, particularly in Layer V of the cortex (Fig. 6F). Many tdTomato⁺ cells were also found in the white matter regions of the corpus callosum, cerebral peduncle, and optic nerve (Fig. 6H). The total number of tdTomato positive cells across the brain was quantified, excluding the choroid plexus as it was not consistently intact in all tissues, but had tdTomato positive cells in the sham and CHI mice (Supplementary Fig. 5A, B). In mice that are tdTomato⁺ but IL-1 β -IRES-Cre-ER^{T2} negative, and in IL-1 β -TRAP-reporter mice that did not receive Tamoxifen, we found no tdTomato-positive cells (Fig. 6I), confirming the selectivity of the IL-1 β -TRAP-reporter mice and the accuracy of our quantification methods.

In IL-1 β -TRAP-reporter mice that received tamoxifen, sparse labeling was detected in sham-injured mice, but this difference was not statistically significant from Cre-negative or naive mice. However, injured mice showed a significant increase in tdTomato-positive cells compared to all other groups (Fig. 6H). While quantifying tdTomato positive cells, it was observed that the vast majority had the morphological appearance of microglia. In injured mice, cells with an astrocyte morphology were occasionally found (Supplementary Fig. 3C). Despite not observing tdTomato-positive cells with a vascular appearance, significant autofluorescence in the vasculature may have obscured our ability to detect these cells. Nonetheless, bright tdTomato⁺ cells near the vasculature were detected, likely perivascular macrophages, suggesting that if there

were vascular IL-1 β -expressing cells, we would have been able to detect them (Supplementary Fig. 3D).

To confirm that tdTomato⁺ cells were microglia, we used confocal microscopy and demonstrated that the majority of tdTomato⁺ cells were Iba-1⁺. After injury, the Iba-1⁺/tdTomato⁺ cells were not necessarily the most hypertrophic (Fig. 6J). In a field of hypertrophic Iba-1⁺ cells, tdTomato cells were among the smaller cells (Fig. 6J). While the production of IL-1 β can lead to cell death via pyroptosis, the presence of tdTomato-positive cells at 6 days post-expression indicates that microglia can produce IL-1 β without undergoing pyroptosis (Evavold et al., Jan. 2018; Sollberger, 2022). However, it is not known if examination at a more proximal time, such as 9–12 h post-injury, would reveal significantly more tdTomato-positive cells, including neutrophils and monocyte-derived macrophages, which may be eliminated within the first day (neutrophils) and first 5 days (monocyte-derived macrophages) (Morganti et al., 2015). Taken together, these data suggest that in pre-clinical models of CNS injury, the IL-1 β -TRAP-reporter can be used to track IL-1 β -expressing cells and allow for direct visualization and future genetic manipulation.

4. Discussion

In this study, we have generated the first mouse line with inducible Cre knocked into the *Il1b* gene. This line allows us to produce an IL-1 β reporter line. We observed that homeostatic IL-1 β expression is restricted to macrophages on the brain's borders and within the ventricles, showing the main source of IL-1 β in the brain are choroid plexus macrophages. Following both peripheral and central inflammatory stimulation, microglia were the major induced IL-1 β expressing cell type in the brain parenchyma; however, this expression was limited to subpopulations of microglia. Long-term tracking of homeostatic IL-1 β expression or of IL-1 β expression under several neuroinflammatory stimuli revealed novel neuronal sources of *Il1b* mRNA, specifically in the entorhinal cortex neurons projecting to the dentate gyrus during homeostasis, in dentate gyrus granule cells during septic neuroinflammation, and in the medium spiny neurons of the nucleus accumbens following chemically induced seizures. This novel mouse line was further utilized in a pre-clinical model of TBI and revealed injury-related IL-1 β -expressing microglia at the site of injury and further demonstrated the effectiveness and robustness of this mouse line.

4.1. Adoption of the cFOS-TRAP mouse model for neuroimmune research

The generation of animal models in order for Targeted Recombination in Active Populations (or TRAP) led to many pivotal discoveries in the neuroscience and neuroimmunology fields. The ability to sensitively and temporally identify contextually active populations of neurons using the Fos-TRAP2 mouse line allowed for the visualization and genetic modification of neural engrams which demonstrated causal relationships between activated cells and their downstream behaviors (DeNardo et al., 2019; Liu et al., 2012; Roy et al., 2022). Further, recent

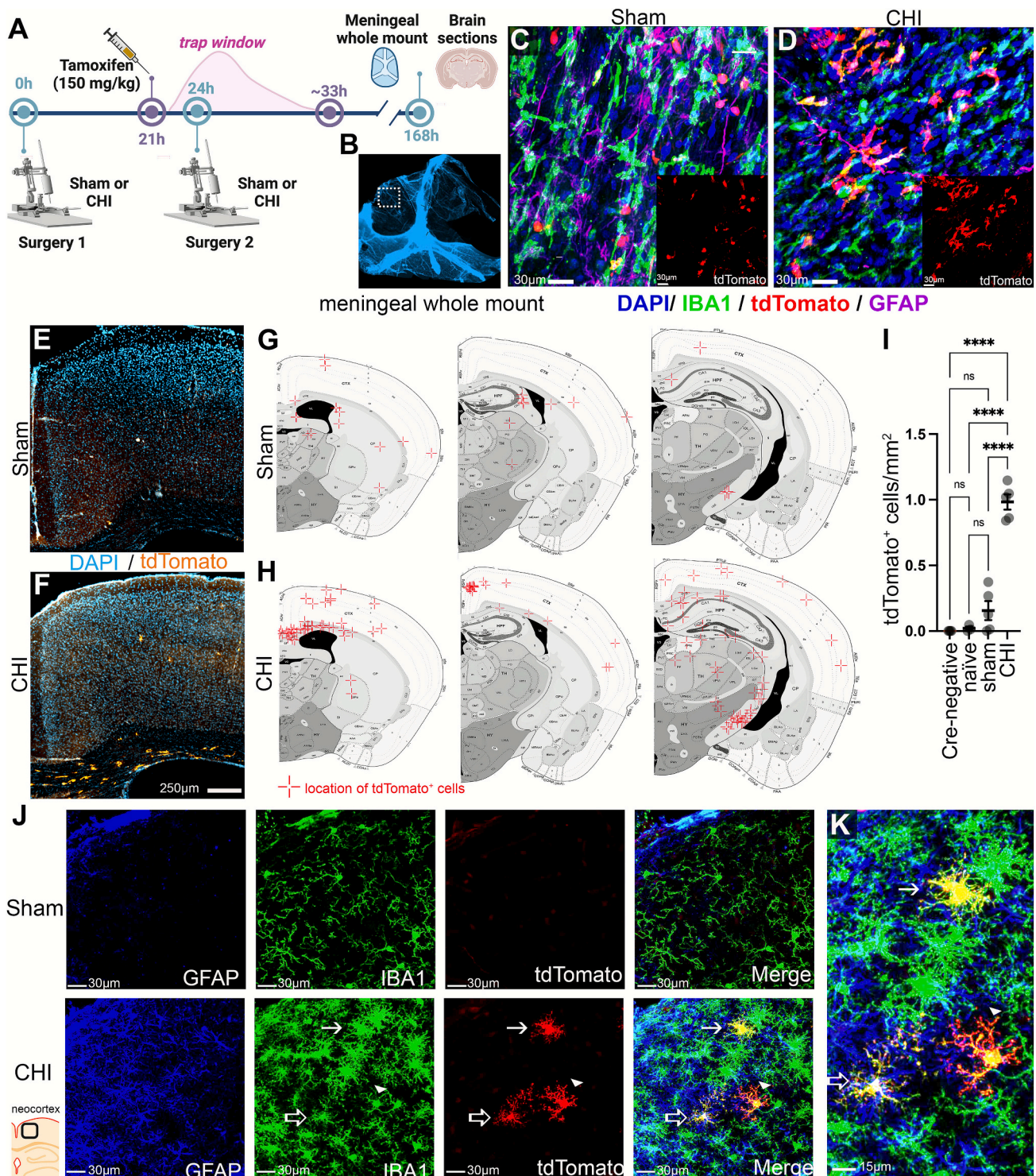


Fig. 6. Analysis of IL-1 β -producing cells in IL-1 β -TRAP-reporter mice following repetitive mild traumatic brain injury (TBI). (A) Experimental design schematic of the two-hit model of repetitive closed head injury (CHI). Mice were given a single injection of Tamoxifen (150 mg/kg i.p.) 3 h before the second CHI surgery. Pink box indicates expected duration of active recombination following a single i.p. injection of tamoxifen. (B) Image showing the locations on the meningeal whole mount where the confocal images in C and D were taken. Representative images of tdTomato positive cells in sham-injured animals show little overlap with Iba-1 or GFAP markers (C), with an increase in size, number, and overlap with Iba-1 in the CHI animals (D). The insets in C and D show the tdTomato channel, demonstrating the few small-sized cells present in the sham compared to the CHI group. (E-F) Immunofluorescent image of DAPI and tdTomato in the cortex of sham-injured (E) animals show sparse labeling and no consistent pattern. (F) This is contrasted with the increased number of tdTomato⁺ cells in CHI-injured mice, with a notable distribution in the cortex under the impact device in Layer V of the cortex and in white matter regions such as the corpus callosum. (G-H) Spatial map of the tdTomato cells in the Sham (G) vs. CHI groups (H). The atlas images represent the 3 slices used to quantify the number of tdTomato⁺ cells per mm² of tissue. The red crosses show the location of counted tdTomato⁺ cells. (I) Quantification of the number of tdTomato⁺ cells across the brain in Cre-negative, Naïve, Sham, and CHI (****p < 0.0001, Tukey test). (J-K) Fluorescent confocal microscopy images of GFAP (blue), Iba-1 (green), tdTomato (red), and merged channels after Sham (J) or CHI (K) surgery. (L) Higher magnification images demonstrating tdTomato⁺ cells in a field of hypertrophic Iba-1⁺ cells, yet two tdTomato⁺ cells (open arrow and arrowhead) were among the smaller hypertrophic cells. Fig. 6A was created in BioRender.

applications of this method to neuroimmunology research identified neural control of immune activity via brain-to-immune communication (Koren et al., 2021). However, this type of methodology has yet to be applied to other biological systems, specifically within the immune system, where like neurons, immune cells transiently express a variety of de novo synthesized cytokines, chemokines, and other inflammatory molecules. Our IL-1 β -TRAP mouse line reinvented the cFOS-TRAP tool for studying key immunologically relevant signaling molecules.

4.2. Role of choroid plexus (ChPlx) macrophages in health and disease

The ChPlx is an important interface between the CNS and periphery in homeostasis, inflammation, and disease. The ChPlx is comprised of a simple cuboidal epithelium bilayer that is a continuous extension of the ventricular ependymal cell layer. Within this bilayer, the ChPlx hosts a vast network of fenestrated capillaries that aid the production of cerebrospinal fluid (CSF) via the epithelial cell plasma filtration. This tightly regulated system is generally referred to as the blood-cerebrospinal fluid barrier (BCSFB). The ChPlx is also host to a variety of immune cells, including multiple macrophage subtypes. This multicellular interface is important for normal brain development since ablation of ChPlx macrophages, in embryonic stages of mice, leads to disrupted formation of vascular networks which is associated with brain hemorrhage in preterm infants (Chen et al., 2024). During inflammation, the ChPlx is important for immune cell trafficking to the brain, alteration of immune cell phenotype, and peripheral inflammation can cause leakage of inflammatory blood proteins into the CSF (Asgari et al., 2017). In human patients with lethal meningitis, the ChPlx harbors large quantities of brain leukocytes and in an LPS-driven mouse model of meningitis, the ChPlx is a major source of recruited peripheral immune cells and epiplexus macrophages aid in the repair of epithelial cells (Xu et al., 2024). In disease, the ChPlx is known to be an important region aiding the development and progression of the mouse model of multiple sclerosis, experimental autoimmune encephalomyelitis (EAE) (Hochstetler and Lehtinen, 2024). These findings highlight the robust functions of the ChPlx in health and disease. Whether IL-1 signaling is a key component in regulating these vital processes has yet to be determined.

4.3. Choroid plexus macrophages are the primary location of basal IL-1 β production

Il1b transcripts have been detected in homeostasis and are known to fluctuate throughout the day in multiple brain regions (Taishi et al., 1997); however, the cellular sources of IL-1 β were never identified and previous studies generally assumed brain IL-1 β production is expressed by parenchymal microglia or neurons. The major shortcoming of previous studies that analyzed *Il1b* mRNA expression from isolated tissue is that the samples were most likely contaminated with meningeal and choroid plexus tissue (Olney et al., 2022). Indeed, our qPCR analysis of *Il1b* mRNA analysis from microdissected ChPlx and other brain regions revealed that the ChPlx is the primary source of *Il1b* mRNA which is further corroborated with our IL-1 β -TRAP-reporter mice. In agreement with our data, scRNAseq data of adult mice shows that immune cells in all ChPlx regions (3rd ventricle, 4th ventricle, and lateral ventricles) have detectable levels of *Il1b* expression in normal conditions (Dani et al., 2021). The presence of IL-1 β ⁺ macrophages in the choroid plexus and surrounding the brain's borders further suggests that IL-1 β produced here may be the major contributor for total brain IL-1 β levels. While the current study did not test whether ventricular IL-1 β production influences IL-1-dependent brain functions, such as sleep, memory, and stress responses, the distribution pattern of basal IL-1 β seen in this study highlights the possibility that non-parenchymal IL-1 β production could be important. The IL-1 β -TRAP mice should be used in follow-up studies to test whether these choroid plexus macrophages may modify normal brain functions.

4.4. IL-1 β ⁺ macrophages at the brains borders

Border associated macrophages (BAM) are highly heterogeneous and comprise many resident populations such as subdural/leptomeningeal macrophages, dural macrophages, stromal choroid plexus macrophages, choroid epiplexus macrophages, and perivascular macrophages (Sun and Jiang, 2024). Each of these populations are ontologically unique and have various renewal capabilities. Ontological studies of brain macrophages conducted by Goldmann et al. highlight a key distinction in the dynamics of central nervous system (CNS) macrophages: while both microglia and perivascular macrophages exhibit slow turnover and microglia are replenished through self-renewal from resident progenitor cells, choroid plexus macrophages demonstrate a more transient nature. These macrophages are relatively short-lived compared to other CNS macrophages and can be partially replenished by peripheral blood-derived cells. However, this turnover is still in a limited capacity since more than 40 % of all embryonic-derived choroid plexus macrophages remain after 45wks of life in mice (Goldmann et al., 2016). This turnover process throughout normal aging, where embryonic macrophages are replaced with blood-derived macrophages, could explain how brain IL-1 β levels increase with age and ultimately alter brain function (Dani et al., 2021). The ontology of IL-1 β -expressing macrophages in the various brain's borders remains to be determined.

4.5. Role of neuronal IL-1 β expression

It has been hypothesized that IL-1 β is produced by neurons and acts as a neurotransmitter or neuromodulator which led to the hypothesis of an "Interleukin-ergic" system in the brain. The first report of neuronal production of IL-1 β was by Lechan et al. in 1989 where immunoreactivity of IL-1 β was found in neuronal-like cells and fibers of the superchiasmatic nucleus, lateral hypothalamus, and in the hippocampus hilus and mossy fibers (Lechan et al., 1990). Further *in situ* hybridization studies revealed *Il1b* mRNA expression in neuronal-like cells of the cortex, throughout the entire neuron cell layers of the hippocampus (DG, CA3, CA2, CA1), and in the cerebellum granular cell of the rat which was further validated, using the same riboprobes, in cultured cerebellar neurons (Bandtlow et al., 1990). Follow-up studies showed that IL-1 β can be produced from cultured rat superior sympathetic ganglia (Freidin et al., 1992); in lumbar dorsal root ganglion (DRG) neurons (Coprav et al., Aug. 2001). In a most recent study, with RNAscope-based *in situ* hybridization, *Il1b* mRNA was found in microglia-depleted mice in the hippocampus suggesting other sources besides microglia (Goodman et al., 2024). Furthermore, cultured human neurons have been shown to express IL-1 β , have active inflammasomes such as NLRP1, and express active IL-1 β converting enzyme, Caspase-1 (Kaushal et al., 2015). In the present study, while some *Il1b*-expressing neurons were observed, these neurons were rarely and stochastically found in the homeostatic brain and/or were restricted to the entorhinal cortex neurons, a region which has not been previously reported. Most of the detected *Il1b*-expressing neurons were found following a severe immune stimulus (i.p. 5 mg/kg LPS) or following excitotoxic, seizure-inducing, kainic acid injection which suggest these cells may be undergoing a yet undetermined cellular process which allows or requires *Il1b* mRNA expression. The literature and present results reveal neuronal sources of IL-1 β expression is possible under specific circumstances.

4.6. Advantages and limitations of IL-1 β -TRAP compared to other methods

Compared to other mice which track or alter IL-1 β expression, the IL-1 β -TRAP mouse has many advantages. Firstly, this is the first genetic technique to detect *Il1b*-expressing cell types in the choroid plexus during homeostasis. Common detection methods such as *in situ* hybridization (Quan et al., 1998) and immunohistochemistry (Shimada and Hasegawa Ishii, 2021) failed to detect the ventricular sources of IL-

IL-1 β in non-stimulated animals, whereas they can detect IL-1 β expression following a large immune stimulus. More advanced techniques, like single nucleus RNA sequencing (scRNAseq), previously revealed that basal *Il1b* expression is present specifically in immune cells of the ChPlx (Dani et al., 2021). Secondly, the IL-1 β -TRAP mice is the only method to gain genetic access to subsets of cytokine expressing cells. We have demonstrated, with the expression of tdTomato, that *Il1b*-expressing cell populations can be genetically accessed and with this access these unique cell networks can potentially be modified with chemogenetics, optogenetics, or conditional gene knockouts. Finally, while other Cre recombinase-based methods exist in cytokine expressing cells, such as IL-17 (Hirota et al., 2011) and IL-2 (Yamamoto et al., 2013), these are based on the cytokine promoter activity and do not allow for timed Cre recombinase activity as the IL-1 β -TRAP mouse permits. The application of tamoxifen or 4-OHT to induce Cre activity allows for currently active populations to be accessed rather than tracking all cytokine expressing cells that were active throughout their entire life.

The IL-1 β -TRAP mouse does have limitations. Firstly, the IL-1 β -TRAP-reporter mouse only reports *Il1b* mRNA expression, not IL-1 β protein expression, and mRNA expression may not simply equate to protein expression. Other IL-1 β reporter mice such as the pIl1b-DsRed mice can be used for tracking IL-1 β protein expression, however, the pIl1b-DsRed mice did not report any homeostatic IL-1 β expression and requires injury (Matsushima et al., 2010); autoimmune activation (Lévesque et al., 2016); or a large immune stimulus (Freria et al., 2020) to visualize IL-1 β production. Secondly, in our hands, the IL-1 β -TRAP mouse requires upwards of 7 days to visualize tdTomato protein expression. We have yet to test whether tdTomato expression can be visualized at a shorter time period after tamoxifen injection; however, other groups that validated TRAP methods use 7 days as this allows for the brightest expression of tdTomato (Guenther et al., 2013). This limitation prevents this tool from accessing short-lived immune cells like the neutrophil, which are well known to express high levels of IL-1 β and are detected with the pIl1b-DsRed mice in various inflammatory contexts (Matsushima et al., 2010). This limitation can be exploited, however, to exclusively access longer lived populations such as brain microglia. While peripheral macrophages are bone marrow derived and turn over within a few days to a month, microglia and brain macrophages are long lived. By waiting for a month following stimulation, peripheral immune cells will turn over and brain-resident IL-1 β -producing microglia/macrophage networks can be selectively accessed and modified. Finally, IL-1 β is expressed in a biologically inactive pro-form and requires cleavage from an active inflammasome to become biologically active. The IL-1 β -TRAP mouse does not detect biologically active IL-1 β and is only a readout of *Il1b* mRNA expression. Other transgenic mice, such as the IL-1 β based Dual Operating Luciferase (IDOL) mice, drive expression of luciferase and partial IL-1 β fusion protein which, when cleaved by the inflammasome, can be measured with a luciferase detection assay. While the IDOL mouse is useful for detecting whether both the IL-1 promoter is used and whether active inflammasomes are present, this tool lacks cellular specificity and employs low resolution imaging. The non-targeted gene insertion method used to create the IDOL mice may have unintended consequences on detecting true IL-1 β expression, and this methodology lacks sensitivity to detect any brain IL-1 β expression even after an immune stimulus (Iwawaki et al., 2015). It is important to note that the IL-1 β -TRAP-reporter permanently tags cells that have expressed IL-1 β at any point when tamoxifen is active. While this allows comprehensive identification of all cells that have released IL-1 β ; it does not capture the quick temporal dynamics of cytokine release across populations. As a result, comparisons to methods that measure transient IL-1 β production should be interpreted cautiously, since the reporter may overrepresent cells that were active earlier relative to those active later.

5. Conclusion

Overall, we have created a new tool for the study of immunologically-active IL-1 β expressing cells in mice, called IL-1 β -TRAP. This study identified new basal expression patterns of brain IL-1 β and lays the groundwork for using this new tool to detail the unique IL-1 β -related cellular and molecular pathways in physiological and pathological contexts.

CCRediT authorship contribution statement

Daniel P. Nemeth: Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization. **Xiaoyu Liu:** Writing – review & editing, Methodology, Conceptualization. **Loretta Chen:** Data curation. **Margaret R. Hawkins:** Formal analysis, Data curation. **Haroon S. Ali:** Formal analysis, Data curation. **Madison G. Lapid:** Formal analysis, Data curation. **Vesal Farsian:** Data curation. **Ashley Kim:** Data curation. **Julianna Saez:** Data curation. **Gabriella Maxey:** Data curation. **Numana Luqman:** Data curation. **Samantha McGovern:** Project administration, Data curation. **Matt Schrier:** Data curation. **Chris Vargas:** Data curation. **Josh St. Juste-Ellis:** Data curation. **Jasmine Yip:** Data curation. **Joris Romain:** Data curation. **Adam D. Bachstetter:** Writing – review & editing, Supervision, Resources, Formal analysis, Data curation, Conceptualization. **Ning Quan:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.bbi.2025.106177>.

Data availability

Data will be made available on request.

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